

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ

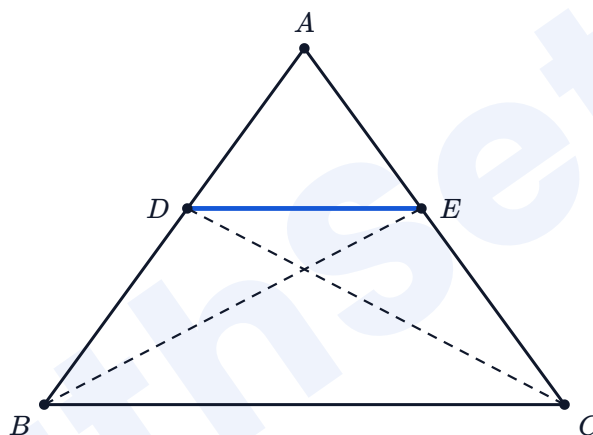
Chapter 6: Triangles

Previous year board questions, topic-wise: 3 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

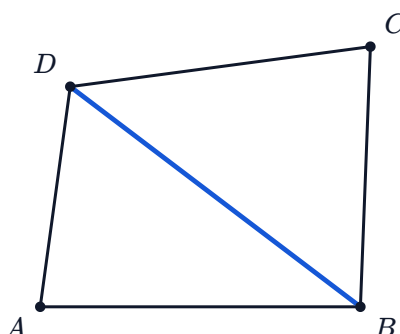
1 Basic Proportionality Theorem

1. Prove that a line drawn parallel to one side of a triangle, cutting the other two sides at distinct points, divides those two sides in the same ratio. **CBSE 2024, 5 marks**

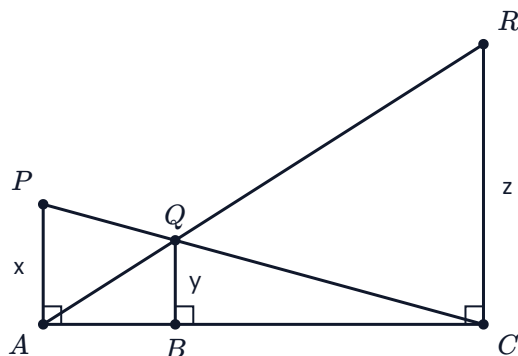


2 Criteria for similarity of triangles

2. In quadrilateral $ABCD$, the diagonal BD bisects both $\angle B$ and $\angle D$. Prove that (i) $\triangle ABD \sim \triangle CBD$ (ii) $AB = BC$. **CBSE 2024, 2 marks**



3. PA , QB and RC are all perpendicular to the line AC , where Q is the point in which PC and AR meet. If $AP = x$, $BQ = y$ and $CR = z$, prove that $\frac{1}{x} + \frac{1}{z} = \frac{1}{y}$. **CBSE 2024, 5 marks**



These questions are based on questions asked in past board examinations. They have been reworded by MathSetu, the figures are redrawn, and the solutions are MathSetu's own. MathSetu is an independent study resource and is not affiliated with or endorsed by CBSE, CISCE, NCERT or any publisher.